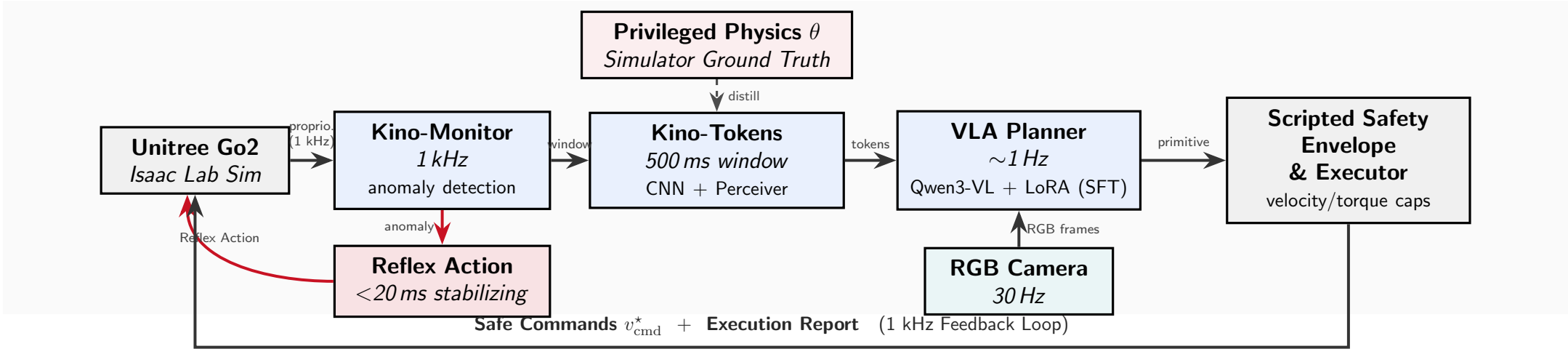
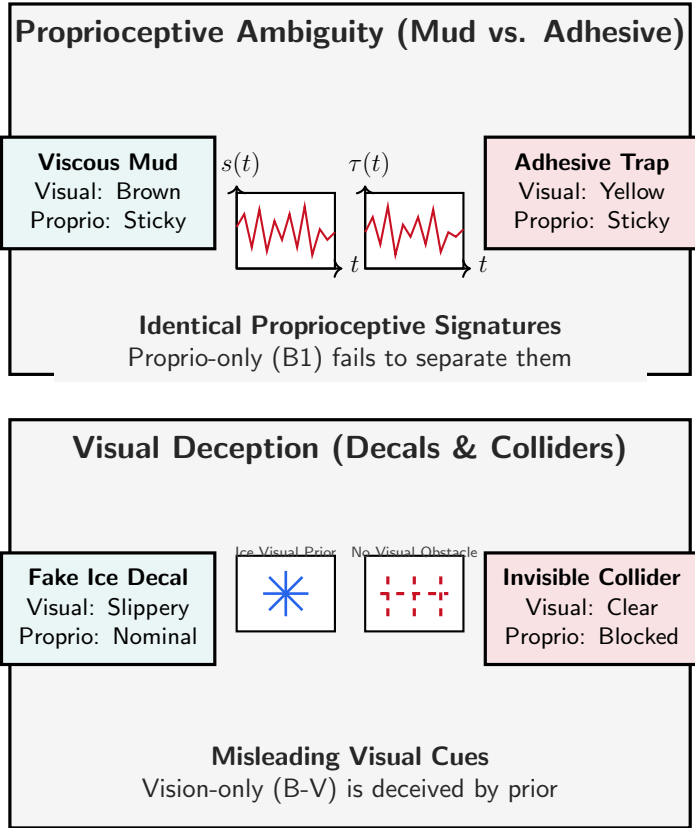


(a) Physical Closed-Loop Pipeline & System Architecture



(b) Cross-Modal Ambiguity & Deception (C1)



(c) Baseline Comparisons on Cross-Modal Conflicts (C2, C5)

Agent	Modality / Architecture	T2 (Proprio Ambiguous)	T3 (Vision Deceptive)	Safety
B1	Pure Proprio (1D-CNN)	✗ Fails	✓ Passes	✗ Unsafe
B-V	Pure Vision (VLM)	✓ Passes	✗ Deceived	✗ Unsafe
B5-unshaped	Naive VLA (No CoT)	✗ Fails	✗ Fails	✗ Unsafe
B5-conflict	Vision-Biased VLA	✓ Passes	✗ Deceived	✗ Unsafe
Ours	KiNO (Bidirectional VLA)	✓ Passes	✓ Passes	✓ Robust

(d) Asymmetric Consequence of Failure Attribution (C3)

